

Data Linking, Shaping, and Cleaning in R

Coumputing Fundamentals for Social Sciences

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Outline

- 1 Introduction
- 2 Creating Example Data
- 3 Data Cleaning
- 4 Data Shaping
- 5 Data Linking (Joining)
- 6 Exercises
- 7 Solutions

What is Data Linking, Shaping, and Cleaning?

- **Data Linking:** combining datasets by common keys.
- **Data Shaping:** changing data format (wide/long).
- **Data Cleaning:** handling missing values, duplicates, and errors.
- Crucial steps before analysis or modeling.

Creating Simple Datasets in R

```
# Dataset 1: Sales data
```

```
sales <- data.frame(  
  ID = 1:6,  
  Product = c("A", "B", "A", "C", "B", "C"),  
  Sales_Q1 = c(100, 150, NA, 200, 130, 210),  
  Sales_Q2 = c(110, 160, 120, NA, 140, 220)  
)
```

```
# Dataset 2: Product info
```

```
products <- data.frame(  
  Product = c("A", "B", "C"),  
  Category = c("Electronics", "Clothing", "Electronics"),  
  Price = c(99.99, 49.99, 79.99)  
)
```

Handling Missing Values

```
# Check missing values in sales
sum(is.na(sales))

# Remove rows with any NA
sales_clean <- na.omit(sales)
```

Handling Duplicates

```
# Add a duplicate row for example
sales_dup <- rbind(sales, sales[1,])

# Remove duplicates
sales_unique <- distinct(sales_dup)
```

Selecting and Filtering

```
library(dplyr)

# Select Product and Sales_Q1 columns
sales_q1 <- select(sales_clean, Product, Sales_Q1)

# Filter rows where Sales_Q1 > 120
high_sales <- filter(sales_clean, Sales_Q1 > 120)
```

Creating New Columns

```
# Calculate total sales Q1 + Q2  
sales_clean <- mutate(sales_clean, Total_Sales = Sales_Q1 +  
  Sales_Q2)
```

Reshaping Data: Wide to Long

```
library(tidyr)

# Convert sales from wide to long format
sales_long <- pivot_longer(sales_clean,
                           cols = starts_with("Sales_Q"),
                           names_to = "Quarter",
                           values_to = "Sales")
```

Reshaping Data: Long to Wide

```
# Back to wide format
sales_wide <- pivot_wider(sales_long, names_from = Quarter,
  values_from = Sales)
```

Joining Sales with Product Info

```
# Join sales and products by Product column
sales_info <- left_join(sales_clean, products, by = "Product")
```

Exercises

- 1 Create the sales and products datasets.
- 2 Check and remove missing values in sales.
- 3 Remove duplicate rows.
- 4 Select sales for Product "A".
- 5 Add a Total_Sales column.
- 6 Reshape sales data to long format.
- 7 Join sales data with product info.

Solutions

```
# See all code in previous slides!  
# Example for filtering Product A:  
sales_a <- filter(sales_clean, Product == "A")
```